

Laplace Transforms

- Any function, $f(t) \longleftrightarrow F(s)$
- Dirac delta or unit impulse, $\delta(t) \longleftrightarrow 1$
- Unit step, $u(t) \longleftrightarrow \frac{1}{s}$
- Unit ramp, $r(t) \longleftrightarrow \frac{1}{s^2}$
- $e^{-at}u(t) \longleftrightarrow \frac{1}{s+a}$
- $\sin(\omega t) \cdot u(t) \longleftrightarrow \frac{\omega}{s^2 + \omega^2}$
- $\cos(\omega t) \cdot u(t) \longleftrightarrow \frac{s}{s^2 + \omega^2}$

Laplace Transform Properties

Linear time-invariant systems with 0 initial conditions

- Linearity: $af_1(t) + bf_2(t) \longleftrightarrow aF_1(s) + bF_2(s)$
- Differentiation: $f^{(n)}(t) \longleftrightarrow s^n F(s)$
- Shifting: $f(t-a)u(t-a), a > 0 \longleftrightarrow e^{-as}F(s)$
- Damping: $e^{-at}f(t) \longleftrightarrow F(s+a)$

Theorems for stable systems

- Initial Value Theorem: $y(0) = \lim_{s \rightarrow \infty} sY(s)$
- Final Value Theorem: $y(\infty) = \lim_{s \rightarrow 0} sY(s)$

Transfer Function, $G(s)$

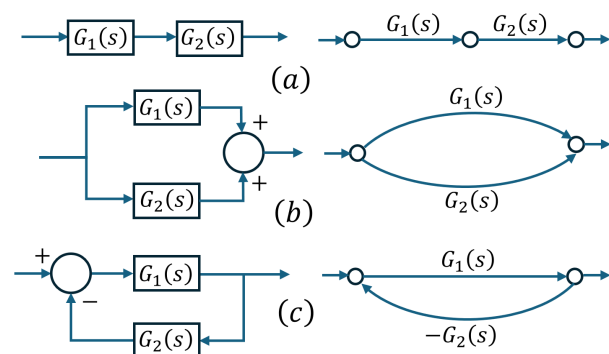
is the ratio of the output, $Y(s)$, to the input, $X(s)$, in the Laplace domain, $G(s) = \frac{Y(s)}{X(s)}$.

System Modeling via Block Diagrams

Three basic building blocks are summer (adder), constant gain, and integrator.

Block diagram/Signal Flow Graph Algebra

Resulting Transfer Function, $G(s)$, for combining two subsystems with transfer functions $G_1(s)$ and $G_2(s)$ in the following configurations: Series (cascade): (a) $\Rightarrow G(s) = G_1(s)G_2(s)$, Parallel: (b) $\Rightarrow G(s) = G_1(s) + G_2(s)$, and Negative feedback: (c) $\Rightarrow G(s) = G_1(s)/(1 + G_1(s)G_2(s))$.



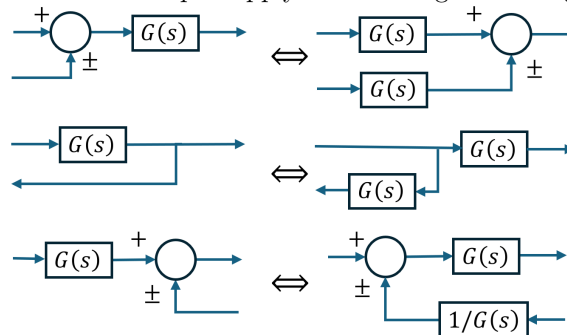
Partial Fraction Decomposition

$$\frac{N(s)}{(s+a)(s+b)} = \frac{A}{s+a} + \frac{B}{s+b}, \quad \frac{M(s)}{(s+a)^2} = \frac{A}{(s+a)^2} + \frac{B}{s+a}$$

- Cross-multiply and compare the numerator terms to solve for unknown variables, A, B , i.e., for the first expression, $N(s) = A(s+b) + B(s+a)$.

Block Diagram Modifications

- Same concepts apply for the signal flow graphs



First-order Systems

If a first-order system transfer function is given by $G(s) = \frac{K}{\tau s + 1}$, then K is the DC Gain or $y(\infty)$ and τ is the time constant, the time it takes for the output to reach 63.2% of $y(\infty)$.

- Rise time (time for $y(t)$ to change from 10% to 90% of $y(\infty)$) is $T_r \approx 2.2\tau$
- Settling times, $T_s(\pm 5\%) \approx 3\tau$ and $T_s(\pm 2\%) \approx 4\tau$

P-Control of First-order Systems

Transfer function of a *stable* plant is $G_p(s) = \frac{K}{\tau s + 1}$ and the controller gain is K_p . The unit step response

- Closed-loop transfer function is $G_{cl}(s) = \frac{K_{cl}}{\tau_{cl}s + 1}$
- Closed-loop system DC gain is $K_{cl} = \frac{KK_p}{1 + KK_p}$
- Closed-loop system time constant is $\tau_{cl} = \frac{\tau}{1 + KK_p}$
- Steady-state error is $E_{ss} = 1 - y(\infty) = \frac{1}{1 + KK_p}$

PD-Control of First-order Systems

Transfer function of a *stable* plant is $G_p(s) = \frac{K}{\tau s + 1}$ and the controller transfer function is $K_p + K_d s$. Closed-loop transfer function is $G_{cl}(s) = \frac{KK_d s + KK_p}{(\tau + KK_d)s + (1 + KK_p)}$.

- Closed-loop DC gain and E_{ss} are similar to P-control.
- Closed-loop system time constant is $\tau_{cl} = \frac{\tau + KK_d}{1 + KK_p}$

Reactive Element $v-i$ Relationships

[**Time-domain**] For a capacitor C , $v'_C = \frac{1}{C}i_C$. For an inductor L , $i'_L = \frac{1}{L}v_L$.

[**Frequency-domain**] For a capacitor C , $\frac{V_C(s)}{I_C(s)} = \frac{1}{sC}$. For an inductor L , $\frac{V_L(s)}{I_L(s)} = sL$.

State-space Representation

- General n -th order ODE, y output and x input: $y^{(n)} + a_{n-1}y^{(n-1)} + \dots + a_1y' + a_0y = b_0x(t)$
- Standard choice of state variables: $x_1 = y, \quad x_2 = \dot{y}, \quad x_3 = \ddot{y}, \quad \dots, \quad x_n = y^{(n-1)}$.
- State-space model in matrix form: $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$ and $y = \mathbf{C}\mathbf{x} + \mathbf{D}u$

- For RLC circuits, state vector is $\mathbf{x} = \begin{bmatrix} v_C \\ i_L \end{bmatrix}$.
- For electric circuits, in general, state vectors are voltages across capacitors and currents through inductors.